

**Contribution of the Late Positive Potential to Emotional Memory: A Joint Neuro-Cognitive
Approach to Memory Prediction**

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Computational modeling of the Early-LPP contribution to emotional recall

The behavioral and EEG analyses identified two forms of variation in recall. At the category level, negative items were recalled more often than neutral items. At the item level, Early LPP was more strongly associated with subsequent recall for negative than neutral items. The empirical analyses distinguish these patterns statistically, but they do not determine how trial-level Early LPP should be mapped onto the encoding mechanism proposed to produce emotional memory enhancement. We therefore tested whether using item-level Early LPP to scale learning strength could replace a category-wide increase in learning strength, or whether the two forms of modulation explained complementary variation in which items were recalled.

We organized this comparison around three focused questions. We tested (1) whether preferential learning of negative items improved prediction of which items were recalled; (2) whether scaling learning with item-level Early LPP could replace that category-wide advantage or improve prediction after the advantage was included; and (3) whether Early-LPP scaling was better applied to both negative and neutral items or only to negative items. We therefore crossed the presence or absence of preferential emotional learning with three LPP specifications: no LPP-based learning modulation, one relationship shared across negative and neutral items, or one relationship limited to negative items. This crossing produced six models. All six models used the same memory architecture. We first describe that shared architecture and show where preferential emotional learning and Early LPP altered learning strength. We then describe model fitting, compare the six models, and use simulations to determine which observed patterns the fitted models reproduced.

eCMR Model Specification

We modeled recall with an adaptation of emotional Context Maintenance and Retrieval (eCMR) (Talmi et al., 2019). Like other retrieved-context models (Howard & Kahana, 2002; Polyn et al., 2009), eCMR represents context as a continuously updated internal state that carries

information from recently processed events. Studying an item changes this state and forms associations in both directions: the item can later reinstate aspects of its study context, and the current context can later provide retrieval support for the item. During free recall, items receiving greater support are more likely to be selected. Recalling an item then reinstates aspects of its study context, changing the cue that guides the next recall. In the present comparison, stronger context-to-item learning could therefore increase an item's probability of entering the recalled set. We first specify this common retrieval mechanism and then define the learning-strength terms that differed across models.

Published eCMR distinguishes temporal and source information within the context representation. Temporal context represents the evolving sequence of recently processed events, whereas source context represents non-temporal information about an event—in the present model, whether an item was negative or neutral. Recalling an item reinstates both components, so its category-related source state can increase support for other items associated with similar source information. Published eCMR also allows preferential learning to strengthen the associations formed by negative items (Talmi et al., 2019). All six models retained this source-representation mechanism: every model represented negative and neutral source information. The comparison varied the learning-strength terms applied within that shared representation, so a model without preferential emotional learning still distinguished negative and neutral items in source context.

The fitted model was an adaptation rather than an unchanged implementation of published eCMR. It omitted item-specific semantic similarities among the pictures and the published source-switch mechanism, in which a transition between negative and neutral categories produces additional disruption of temporal context. We omitted that mechanism because recall order was unavailable to diagnose category-boundary effects and because the present comparison concerned learning strength rather than temporal disruption at source transitions.

The implementation also differed from published eCMR in how it combined old and new context and in how it converted retrieval support into recall choices. We identify those differences when the corresponding operations are introduced below. These choices were fixed across all six

model definitions, so their structural differences concerned the learning-strength terms rather than different retrieval architectures.

Within each list j , each studied item i occupied a unique one-hot feature slot $\mathbf{f}_i$: one entry was 1 and all remaining entries were zero. The complete context state therefore contained the temporal and source components introduced above,

$$\mathbf{c}_i = \begin{bmatrix} \mathbf{c}_i^T \\ \mathbf{c}_i^S \end{bmatrix}.$$

The two components were updated and normalized separately, but together formed one context representation rather than independent memory systems. Both components included a start-of-list state, and all context states and associations were initialized anew for each list.

Items and context interacted through two associative mappings with opposite directions. With column-vector notation, the complete feature-to-context mapping M^{FC} transformed an item into contextual input, whereas the complete context-to-feature mapping M^{CF} transformed the current context into support for candidate items:

$$M^{FC} = \begin{bmatrix} M_T^{FC} \\ M_S^{FC} \end{bmatrix}, \quad M^{CF} = \begin{bmatrix} M_T^{CF} & M_S^{CF} \end{bmatrix}.$$

Core symbols are collected in Supplementary Table 6.

Before study, the two associative mappings were initialized differently. The temporal block of the initial feature-to-context mapping associated each item with its own temporal feature, and the source block associated each item with its negative or neutral source feature; both had weight $1 - \gamma$. In the initial temporal context-to-feature block, the context feature paired with an item supplied support δ , whereas every other item-associated context feature supplied background support α . The source context-to-feature block began at zero, so source context supported items only through associations learned during study.

When an item was presented, each context component first received unit-length input from the corresponding feature-to-context block:

$$\mathbf{u}_i^T = \mathcal{N}\left(M_T^{FC} \mathbf{f}_i\right), \quad \mathbf{u}_i^S = \mathcal{N}\left(M_S^{FC} \mathbf{f}_i\right),$$

where $\mathcal{N}(\mathbf{v}) = \mathbf{v} / \sqrt{\sum_k v_k^2 + \epsilon}$ and ϵ is machine epsilon. Each preceding context state then moved toward its input. Smaller encoding-drift values retained more of the preceding state; larger values gave the current item greater influence.

The fitted implementation performed this movement with the following componentwise integration operator:

$$\begin{aligned} \rho(\mathbf{c}, \mathbf{u}, \beta) &= \sqrt{\mathbf{1} + \beta^2 [(\mathbf{c} \odot \mathbf{u})^2 - \mathbf{1}]} - \beta(\mathbf{c} \odot \mathbf{u}), \\ \mathcal{U}(\mathbf{c}, \mathbf{u}, \beta) &= \mathcal{N}[\rho(\mathbf{c}, \mathbf{u}, \beta) \odot \mathbf{c} + \beta \mathbf{u}]. \end{aligned}$$

Here, $\odot$, the square, and the square root operate elementwise. The retention vector ρ controlled how much of each dimension of the preceding state remained after the new input was incorporated. Temporal and source context were then updated separately:

$$\mathbf{c}_i^T = \mathcal{U}(\mathbf{c}_{i-1}^T, \mathbf{u}_i^T, \beta_{\text{enc}}^T), \quad \mathbf{c}_i^S = \mathcal{U}(\mathbf{c}_{i-1}^S, \mathbf{u}_i^S, \beta_{\text{enc}}^S).$$

This componentwise retention-and-renormalization rule differs from the scalar retention coefficient in the published CMR and eCMR equations.

Both context components were updated before the item's associations were learned. The item was therefore bound to temporal and source states that it had already helped create, rather than to the states that preceded its presentation. This post-update order was held fixed across the comparison.

Feature-to-context learning stored the updated state that a later presentation or recall of the item could reinstate. The same fitted rate γ applied to the temporal and source blocks:

$$\Delta M_{T,i}^{FC} = \gamma \mathbf{c}_i^T \mathbf{f}_i^T, \quad \Delta M_{S,i}^{FC} = \gamma \mathbf{c}_i^S \mathbf{f}_i^T.$$

Context-to-feature learning stored the reverse associations through which context could later support the item:

$$\Delta M_{T,i}^{CF} = \phi_i \mathbf{f}_i(\mathbf{c}_i^T)^\top, \quad \Delta M_{S,i}^{CF} = S_{ij}^S \mathbf{f}_i(\mathbf{c}_i^S)^\top.$$

The temporal learning strength $\phi_i = 1 + \phi_s \exp[-\phi_d(i-1)]$ gave early list positions a temporary advantage: ϕ_s controlled its size and ϕ_d controlled its decay. The source learning strength S_{ij}^S contained the same primacy gradient together with the emotional-learning and Early-LPP terms defined below. Placing both focal predictors on source context-to-item learning gave them one directly comparable route to recall, while temporal context-to-item learning remained unchanged.

At recall onset, β_{start} shifted both context components toward their start-of-list states. The current temporal and source states then supplied additive support for each item that had not yet been recalled:

$$a_i = [M^{CF} \mathbf{c}]_i = [M_T^{CF} \mathbf{c}^T]_i + [M_S^{CF} \mathbf{c}^S]_i.$$

The choice-sensitivity parameter τ_c converted support into choice weights proportional to $a_i^{\tau_c}$. The implementation evaluated these powers in log space and rescaled all weights by a common constant to avoid numerical overflow; this rescaling preserved their relative values. After machine epsilon was added to each available item's weight, let q_i denote the resulting positive value. The probability of the next recalled item was

$$P(R_t = i \mid \mathcal{A}_t) = \frac{q_i}{\sum_{k \in \mathcal{A}_t} q_k}, \quad i \in \mathcal{A}_t,$$

where $\mathcal{A}_t$ is the set of studied items still available at recall step t . Larger τ_c made the model more likely to select the item with greater support.

After an item was recalled, temporal context moved toward its retrieved temporal input at rate β_{rec}^T , and source context fully moved toward its retrieved source input by fixing $\beta_{\text{rec}}^S = 1$. The retrieved source input contained both the item's category feature and the source state learned during study.

Full source reinstatement was a fixed modeling constraint, not a conclusion from the data: without recall order, we treated partial source reinstatement as difficult to distinguish from source-learning strength and recall-choice sensitivity. The recalled item was then removed from the available set, so repeated recalls were disallowed. Context-guided selection and reinstatement complete the pathway through which source learning strength can change which items enter the recalled set.

Encoding-strength hypotheses

Every model estimated a shared source-learning baseline $\omega\phi_i$. The parameter ω scaled source learning relative to temporal learning, for which the corresponding scale was fixed at 1. Estimating ω in every model allowed source context to contribute even when neither focal mechanism was enabled. The first focal term allowed negative items to receive a category-wide emotional-learning multiplier on this shared baseline.

Preferential emotional learning was represented by $\phi_{\text{emot}}^{e_i}$, where $e_i = 1$ for a negative item and $e_i = 0$ for a neutral item. The multiplier therefore equaled ϕ_{emot} for negative items and 1 for neutral items. Models without the preferential emotional-learning term fixed $\phi_{\text{emot}} = 1$, whereas models testing preferential emotional learning estimated the multiplier over the range (0, 10]. Values above 1 strengthened source context-to-item learning for negative items relative to neutral items, whereas values below 1 attenuated it. That range allowed attenuation as well as enhancement of negative-item learning; “preferential emotional learning” names the directional hypothesis, and interpreting the fitted term as enhancement requires an estimate above 1. The category-wide emotional-learning multiplier can change the average recall difference between negative and neutral items, but it cannot distinguish two negative items on the basis of their different Early-LPP amplitudes. The Early-LPP terms supplied that item-level variation.

For item i on 20-item study-test list j , the model used within-list mean-centered Early LPP,

$$z_{ij} = \text{EarlyLPP}_{ij} - \overline{\text{EarlyLPP}_j}.$$

A positive value of z_{ij} means that an item elicited a larger Early LPP than the average item

on the same list; a negative value means that it elicited a smaller one. This predictor isolates within-list variation and removes differences between list means by construction. The present models can therefore test whether relatively larger Early LPP predicts which items enter a list's recalled set, but not whether lists with larger average Early LPP produce more recalls. The preceding empirical regression did not separate within-list variation from differences among list means and participants. The computational coefficient addresses only the within-list relationship and is therefore not a numerical counterpart of the empirical regression coefficient.

Source learning strength was the product of the ordinary source-learning baseline, the category-wide emotional-learning multiplier, and an item-level Early-LPP multiplier:

$$S_{ij}^S = \omega \phi_i \times \phi_{\text{emot}}^{e_i} \times \exp[\kappa z_{ij} + \kappa_{\text{emot}} e_i z_{ij}] .$$

The coefficient κ is the General Early-LPP slope on log learning strength, and κ_{emot} is an additional negative-item slope. The resulting log-learning slope is κ for neutral items and $\kappa + \kappa_{\text{emot}}$ for negative items. At the list mean, where $z_{ij} = 0$, the Early-LPP multiplier equals 1 and the model recovers its ordinary category-level learning baseline. The exponential link guarantees positive learning strength, but positivity is not its only implication: the link assumes that a one-unit Early-LPP difference produces a proportional rather than a constant absolute change in learning. The model comparison tests whether one such proportional relationship improves prediction when it applies to both categories or only to negative items.

No-LPP models fixed both Early-LPP coefficients at zero. General-LPP models estimated κ and fixed $\kappa_{\text{emot}} = 0$, applying one slope to negative and neutral items. Emotion-dependent-LPP models fixed $\kappa = 0$ and estimated κ_{emot} , applying one slope only to negative items. General and Emotion-dependent models therefore estimated the same number of parameters and differed only in the items to which their single Early-LPP slope applied. The empirical interaction motivated this matched comparison, but the model comparison determined which mapping better predicted the recalled sets.

Model-comparison design

Crossing a fixed or freely estimated emotional-learning multiplier with the three Early-LPP mappings produced the six models in Table 1.

Canonical implementation identifiers corresponding to these manuscript labels are reported in Supplementary Table 5.

The grid supports the three comparisons introduced at the beginning of this section. Within each row, comparing the two columns tests whether freeing the emotional-learning multiplier improves prediction while holding the Early-LPP mapping fixed. Within each column, comparing an Early-LPP model with the No-LPP model tests whether the neural measure adds predictive information while holding fixed whether the emotional-learning multiplier was estimated. Finally, comparing General with Emotion-dependent LPP within either column tests two equal-complexity placements of one Early-LPP coefficient. The shared architecture in every cell makes these contrasts interpretable.

The grid also implements the substitution test proposed in the original motivation for constraining encoding strength with LPP. The General-LPP model with the emotional-learning multiplier fixed and the Preferential-learning model without LPP each had 12 free parameters. Both models added one scaling term to the same source context-to-item learning strength: the General-LPP model scaled learning from item-level LPP, whereas the Preferential-learning model scaled negative-item learning with ϕ_{emot} . Their comparison therefore tests whether LPP-derived scaling can replace a free emotional-learning multiplier; the equal-complexity Emotion-dependent-LPP model provides a second LPP-derived alternative.

Table 2 summarizes the parameters and the constraints that created the six cells. The eleven common free parameters governed temporal and source updating, pre-experimental support, associative learning, primacy, and recall choice. Estimating the emotional-learning multiplier or either one-slope Early-LPP mapping added one free parameter.

The primary grid does not estimate General and emotion-dependent Early-LPP coefficients simultaneously. It therefore asks whether one coefficient is more useful when shared

by both categories or limited to negative items; it does not test whether a residual General component is exactly zero after an emotion-dependent component is fitted. The same learning-strength equation also defines a Full-LPP mapping in which κ and κ_{emot} are estimated together. In that mapping, κ_{emot} is formally an additional negative-item slope; when $\kappa = 0$, it supplies the entire negative-item slope. We fitted the two source-context Full-LPP variants exploratorily, but their best-found solutions did not consistently recover the likelihoods of their nested one-slope models. We therefore excluded them from the primary comparison and report them separately as optimization-incomplete fits.

The comparison also holds source context constant across all six models. It can test learning-strength terms within that architecture, but it cannot establish that the predictive Early-LPP relationship is specific to source-context rather than temporal-context learning. Having defined the alternatives, we next describe the common evidence used to compare them.

Pooled estimation and evaluation

The modeling dataset matched each of the 6,840 study events to its recall outcome and Early-LPP value. As described in the preceding EEG methods, each single-trial ERP was first standardized relative to its own prestimulus interval, using the mean and standard deviation from -200 to 0 ms separately for each electrode. Early LPP was then extracted from the prespecified 400 – 1000 ms window and centroparietal electrodes.

Early-LPP values were unavailable for 371 events (5.4%). Before within-list centering, these values were imputed in three steps: first from available items of the same condition on the same participant’s list, then from all available items on that list if the condition-specific mean was unavailable, and finally from the dataset-wide mean for the corresponding condition. Recall status was not used in the imputation. This procedure yielded one complete 20-item input sequence for each list.

Each model was fitted with one parameter vector shared across all 342 study-test lists from 38 participants. The pooled analysis asks whether one parameterization captures the population-level pattern across the full dataset. It does not estimate participant-specific model

parameters or individual differences. This restriction is important because each participant contributed only nine lists, too few independent list histories to support stable estimation of 11 to 13 parameters per participant.

The available recall records identified which items were recalled, but their output order was not available for modeling. This creates a problem because eCMR assigns probability to an ordered recall sequence: after each recall, the retrieved item changes context and therefore changes the probabilities of subsequent recalls. An unordered set of n_j items is compatible with $n_j!$ possible orders. Enumerating all of those orders was impractical, so we approximated the evidence for each recalled set using 50 sampled orders.

Let $\mathcal{S}_j$ denote the items recalled from list j , let $n_j = |\mathcal{S}_j|$, let π_{jk} be the k th sampled ordering of those items, and let θ denote a model's parameter vector. For each list, the model was supplied with n_j and evaluated on which studied items composed $\mathcal{S}_j$. Using $K = 50$ fixed sampled orders, the fitted score for the set was

$$\tilde{P}(\mathcal{S}_j \mid \theta, n_j) = \frac{1}{K} \sum_{k=1}^K \prod_{t=1}^{n_j} P(R_{jt} = \pi_{jk,t} \mid \pi_{jk,1:(t-1)}, \theta) .$$

Each product in this expression is the probability of one possible ordered recall sequence; the outer average combines the 50 sampled sequences. Multiplying the average by $n_j!$ would estimate the full sum over compatible orders. Because this factor depends only on the observed list and not on the model parameters, omitting it adds the same constant to every negative log-likelihood (NLL). It therefore does not change fitted parameters; after model complexity is considered, it likewise does not change differences or rankings under Akaike's information criterion (AIC), defined below. The same sampled orders and random seed were used for every model, so differences between models could not arise from evaluating different order samples.

Termination events were excluded from the likelihood. Accordingly, the model predicts which items enter the recalled set conditional on the number of items the participant recalled from that list. The analysis does not explain why that number of items was recalled and does not test a process for deciding when recall stops. Stopping may depend on the state of retrieval in free

recall, but evaluating such a process requires a separate, validated termination model. The current likelihood keeps the present comparison focused on recalled-item identity.

We minimized the summed negative log of the approximate set scores using differential evolution, a population-based parameter search ([Storn & Price, 1997](#)). Each optimization began with a Latin-hypercube sample that spread its initial candidate vectors across the allowed parameter space. The search population contained 15 candidate vectors per free parameter, yielding 165, 180, or 195 candidates for models with 11, 12, or 13 free parameters, respectively. The crossover probability was 0.9, the differential weight varied between 0.5 and 1.0, and the relative convergence tolerance was 10^{-4} . Optimization continued for at most 1,000 generations and was repeated from three independent starts; the lowest NLL found across the three starts was retained.

The Early-LPP coefficients were constrained to be nonnegative in accordance with the directional empirical hypothesis. Their upper bound of 0.2145 limited either Early-LPP term to a tenfold multiplier at the largest observed centered Early LPP. The resulting estimates and rankings are conditional on this search budget and these parameter ranges.

Lower NLL indicates that a model assigned greater probability to the observed recalled sets. We compared the best-found solutions using Akaike's information criterion, $AIC = 2NLL + 2p$, where p is the number of free parameters ([Burnham & Anderson, 2002](#)). Lower AIC indicates a better trade-off between conditional fit and model complexity. AIC establishes whether an optional learning term improved prediction enough to offset its additional parameter, but AIC does not show which empirical pattern that term reproduced. We therefore paired the model comparisons with simulations designed to make the fitted terms interpretable.

For each fitted parameter vector, we generated 200 complete simulated datasets containing the same 342 study lists and the same item-level Early-LPP values as the observed dataset. For each list, we generated exactly its observed number of recalled items, matching the conditioning used in the likelihood.

We summarized two patterns in each simulated dataset: the negative-versus-neutral recall

difference and the remembered-versus-forgotten difference in within-list centered Early LPP. The parameters changed which studied items the model recalled; they did not change the observed Early-LPP value attached to an item. These summaries were not optimized as separate fitting targets. They are therefore explanatory diagnostics of the fitted recalled-set model, not independent or out-of-sample validation.

Model-comparison results

Overall comparison

The model combining preferential emotional learning with Emotion-dependent Early-LPP modulation had the lowest AIC (Table 3). Its best-found NLL was 8240.310 with 13 free parameters, yielding an AIC of 16506.619. The overall ranking provides an initial orientation; the three planned comparisons determine what the ranking means. We report those comparisons in the order introduced above.

The fit statistics show whether a term improved conditional prediction, whereas the simulations show what empirical pattern the fitted term produced. Figure 1 displays the observed benchmarks and four predictions selected from the six models to isolate the three planned contrasts. For preferential emotional learning, the figure compares the Emotion-dependent-LPP model with and without a free emotional-learning multiplier. For the added contribution of Early LPP, it compares preferential emotional learning alone with a model that also included Emotion-dependent LPP. For the placement of the Early-LPP slope, it compares models that combined preferential emotional learning with either General or Emotion-dependent LPP. The left panels show the category-level recall pattern. The right panels show the item-level Early-LPP pattern separately for negative and neutral items. All six models remain in the numerical tables even though four representative predictions are sufficient to display the contrast logic.

Contribution of preferential emotional learning

The first comparison asks whether preferential learning of negative items improves prediction of which items are recalled. Freeing the emotional-learning multiplier lowered AIC under every Early-LPP specification: by 113.639 points with no LPP, by 97.238 points with

General LPP, and by 78.768 points with Emotion-dependent LPP. The fitted multiplier exceeded 1 in all three models, so the estimated term enhanced rather than attenuated negative-item learning. The recall-rate panels explain this consistent advantage. Models with preferential emotional learning produced negative-minus-neutral recall differences between 0.118 and 0.134, close to the observed difference of 0.124; models without preferential emotional learning produced differences between 0.001 and 0.019. Because every model represented negative and neutral source features, this result concerns preferential emotional learning rather than the mere presence of category information. Preferential emotional learning supplied category-level recall structure, leaving the item-level contribution of Early LPP as the next question.

Contribution of Early LPP

The second comparison asks whether Early LPP adds predictive information beyond a model with no EEG input. Without preferential emotional learning, General LPP lowered AIC by 23.582 points and Emotion-dependent LPP lowered it by 63.676 points relative to the source-context baseline. With preferential emotional learning, the corresponding AIC improvements were 7.182 and 28.805 points. Early LPP therefore improved conditional prediction when the emotional-learning multiplier was fixed and when it was estimated.

The equal-complexity substitution comparisons asked a different question and produced a different result. The Preferential-learning model had an AIC 90.057 points lower than the General-LPP model and 49.963 points lower than the Emotion-dependent-LPP model; all three models had 12 free parameters. Within the fitted specification and parameter ranges, neither LPP-derived scaling rule matched the free emotional-learning multiplier, even though Early LPP improved prediction when added to models with or without that multiplier. This pattern favors complementarity over substitution.

The Early-LPP panels identify the additional pattern captured by Early-LPP modulation. In the observed data, remembered items had a 0.403 larger mean centered Early LPP than forgotten items within the negative category, whereas the corresponding neutral-item difference was 0.021. Models without Early-LPP modulation produced little remembered-versus-forgotten

separation in either category. Because the Early-LPP values themselves remained fixed, remembered–forgotten differences arose only when a model preferentially recalled items with larger values. Early-LPP modulation therefore supplied item-level information that the category-wide emotional-learning multiplier did not contain. Because both Early-LPP mappings added information, the final comparison asks where their single slope should apply.

General versus Emotion-dependent Early LPP

The third comparison contrasts one slope shared across categories with one slope limited to negative items. The two alternatives had equal numbers of parameters, so their AIC difference reflected fit rather than a difference in the complexity penalty. Emotion-dependent LPP lowered AIC by 40.094 points relative to General LPP when preferential emotional learning was absent and by 21.623 points when preferential emotional learning was present. The simulations clarify why: General LPP separated remembered and forgotten items in both categories, whereas Emotion-dependent LPP concentrated the separation among negative items, as in the observed pattern. When the models were each given one Early-LPP coefficient, that coefficient was therefore more useful when applied to negative items.

This comparison is conditional on the declared parameter ranges. In the model combining preferential emotional learning with General LPP, $\hat{\kappa} = 0.208$, or 97% of its upper bound of 0.2145, and no wider-bound sensitivity analysis was conducted. We therefore cannot establish whether the Emotion-dependent mapping would remain preferred under a wider Early-LPP bound. The supported conclusion is narrower: within the declared range and equal-complexity comparison, the Emotion-dependent mapping predicted the recalled sets better.

A second near-bound estimate cautions against interpreting the General-LPP coefficients in isolation. In the General-LPP fit with the emotional-learning multiplier fixed, the source-learning scale approached its upper bound: $\hat{\omega} = 9.047$ on the range $(0, 10]$. This pattern suggests that the source-learning scale and the General Early-LPP coefficient may compensate for one another.

The focal fitted parameters in Table 4 show that the preferred model used both proposed mechanisms. The preferred model, which combined preferential emotional learning with

Emotion-dependent LPP, estimated a source-learning scale of 0.930, an emotional-learning multiplier of 1.946, and an emotion-dependent log-learning slope of 0.128. The source-learning scale, context drift, and choice sensitivity were estimated jointly with these terms, so isolated parameter magnitudes should not be assigned direct psychological meanings or compared across models without considering the complete parameter vector.

The exploratory Full-LPP fits did not resolve whether a residual General component remained. Their original best-of-three solutions failed a basic optimization check: their NLLs were 3.265 and 1.294 points above their nested Emotion-dependent models without and with preferential emotional learning, respectively. Evaluating each nested parameter vector through the corresponding Full likelihood at $\kappa = 0$ reproduced the parent NLL exactly. We therefore excluded the Full variants from the six-model ranking, report both the original and embedded-parent evaluations in Supplementary Table 9, and do not use them to infer whether a residual General component remains.

Summary and scope

Taken together, the comparisons support complementary roles for preferential emotional learning and Early-LPP modulation. The emotional-learning multiplier reproduced the higher recall rate for negative than neutral items. The item-level Early-LPP multiplier distinguished which negative items were recalled. The preferred specification combined preferential emotional learning with Emotion-dependent LPP. It implements the proposed pathway from Early LPP through source context-to-item learning strength to the conditional probability of entering the recalled set, while also reproducing the emotional-memory advantage. The comparison thereby implements the original proposal to use an independent neural measure to constrain the same source-learning pathway governed at the category level by ϕ_{emot} . The results show that LPP-derived scaling complemented, rather than replaced, the fitted emotional-learning multiplier.

The conclusion is deliberately narrower than a general account of emotional memory. It concerns within-list variation in Early LPP and recalled-item identity conditional on the observed recall count, using one parameter vector shared across participants.

The analysis does not explain total recall or stopping and does not estimate effects of between-list mean Early LPP. It also does not characterize participant-specific mechanisms, compare source with temporal implementations, or identify Early LPP uniquely with attention, arousal, or stimulus significance. Those questions require different data or additional models rather than stronger interpretation of the present comparison.

Finally, the numerical conclusions are conditional on the 50-order recalled-set approximation, the declared parameter bounds, and the differential-evolution search. The mechanistic conclusions concern the stated eCMR adaptation, including its componentwise context update and its omission of semantic similarity and source-switch disruption. Within those limits, a category-wide emotional-learning multiplier and an emotion-dependent item-level neural multiplier accounted for complementary structure in the observed recalled sets.

Supplementary material

Model-label mapping

Core notation

Table 6 collects the formal objects that recur across the specification. The fitted parameters and their roles are listed separately in Table 2.

Complete pooled parameter estimates

Table 7 defines the compact model codes used in Table 8.

Exploratory Full-LPP fits

Table 9 reports the original optimizer outputs and the explicitly evaluated nested-parent candidates. Because a model cannot fit worse than a nested model at the optimum, the nesting violations indicate incomplete optimization rather than evidence that the General slope harmed fit. With the same population-size factor used in the primary analysis, the 13- and 14-parameter Full models used 195 and 210 candidate vectors, respectively. The embedded-parent candidate was the best evaluated Full-LPP solution in each case, and its AIC continues to count the additional Full-model parameter. Neither candidate is treated as a demonstrated maximum-likelihood

solution.

Table 10 reports the original optimizer parameter estimates. These values are retained for transparency, not interpreted individually. Parameter roles and fitted ranges are defined in Table 2.

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Table 1

Six models formed by crossing Early-LPP modulation with a fixed or estimated emotional-learning multiplier. All models contained temporal and negative/neutral source-context components; parenthetical values are numbers of free parameters. Estimating the multiplier tested the directional hypothesis of preferential emotional learning while also permitting attenuation.

Early-LPP modulation	Emotional-learning multiplier	Emotional-learning multiplier
	fixed at 1	estimated
None	Source-context baseline (11)	Preferential-learning model (12)
General	General-LPP model (12)	Preferential learning + General-LPP model (13)
Emotion-dependent	Emotion-dependent-LPP model (12)	Preferential learning + Emotion-dependent-LPP model (13)

Table 2

Parameters in the six-model specification. Bounds shown as open at zero were implemented with machine epsilon as the positive lower bound.

Parameter	Role	Fitted range or fixed value
β_{enc}^T	Temporal-context drift during study	(0, 1)
β_{enc}^S	Source-context drift during study	(0, 1)
β_{start}	Drift toward start-of-list context before recall	(0, 1)
β_{rec}^T	Temporal-context reinstatement after recall	(0, 1)
β_{rec}^S	Source-context reinstatement after recall	1 (fixed)
α	Shared pre-experimental context-to-item support	(0, 100]
δ	Item-specific pre-experimental context-to-item support	(0, 100]
γ	Feature-to-context learning rate	(0, 1)
ϕ_s	Primacy scale	(0, 100]
ϕ_d	Primacy decay	(0, 100]
τ_c	Recall-choice sensitivity	(0, 100]
ω	Source learning relative to temporal learning	(0, 10]
ϕ_{emot}	Emotional-learning multiplier for negative items	1 (fixed) or (0, 10]
κ	General Early-LPP slope on log learning strength	0 (fixed) or [0, 0.2145]
κ_{emot}	Emotion-dependent Early-LPP	0 (fixed) or [0, 0.2145]

Table 3

Pooled fit of the six eCMR model variants. p is the number of free parameters; NLL is the best-found negative log-likelihood across three differential-evolution starts; ΔAIC is relative to the lowest-AIC model.

Model	p	NLL	AIC	ΔAIC
Source-context baseline	11	8313.531	16649.063	142.443
General-LPP model	12	8300.740	16625.480	118.861
Emotion- dependent-LPP model	12	8280.693	16585.387	78.768
Preferential- learning model	12	8255.712	16535.424	28.805
Preferential learning + General-LPP model	13	8251.121	16528.242	21.623
Preferential learning + Emotion- dependent-LPP model	13	8240.310	16506.619	0.000

Table 4

Fitted source-learning scale and parameters governing preferential emotional learning and Early-LPP modulation. Fixed entries are the values that disable the corresponding optional mechanism.

Model	ω	ϕ_{emot}	κ	κ_{emot}
Source-context baseline	0.226	1 (fixed)	0 (fixed)	0 (fixed)
General-LPP model	9.047	1 (fixed)	0.060	0 (fixed)
Emotion- dependent-LPP model	1.385	1 (fixed)	0 (fixed)	0.184
Preferential- learning model	1.452	1.504	0 (fixed)	0 (fixed)
Preferential learning + General-LPP model	1.037	7.525	0.208	0 (fixed)
Preferential learning + Emotion- dependent-LPP model	0.930	1.946	0 (fixed)	0.128

Table 5*Mapping between the manuscript-facing labels and the executable model registry.*

Manuscript label	Canonical registry identifier
Source-context baseline	CategoryOnly_eCMR
General-LPP model	CategoryOnly_eCMR_LPP_General
Emotion-dependent-LPP model	CategoryOnly_eCMR_LPP_EmotionalOnly
Preferential-learning model	EEM_eCMR
Preferential learning + General-LPP model	EEM_eCMR_LPP_General
Preferential learning + Emotion-dependent-LPP model	EEM_eCMR_LPP_EmotionalOnly

Table 6*Core notation used in the model specification and recalled-set likelihood.*

Symbol	Meaning
i, j, t	Studied item, study-test list, and recall-output position, respectively
$\mathbf{f}_i$	One-hot feature vector identifying studied item i
$\mathbf{c}, \mathbf{c}^T, \mathbf{c}^S$	Complete context state and its temporal and source components
$\mathbf{u}_i^T, \mathbf{u}_i^S$	Temporal and source contextual input generated by item i
M^{FC}, M^{CF}	Complete feature-to-context and context-to-feature associative mappings
$M_T^{FC}, M_S^{FC}, M_T^{CF}, M_S^{CF}$	Temporal and source blocks of the two associative mappings
ϕ_i	Serial-position-dependent primacy multiplier for item i
S_{ij}^S	Completed source context-to-item learning strength for item i on list j
e_i	Negative-item indicator: 1 for negative and 0 for neutral
z_{ij}	Within-list mean-centered Early LPP for item i on list j
a_i, q_i	Context-derived support for item i and its positive choice weight
$\mathcal{A}_t$	Studied items still available at recall position t
S_j, n_j	Observed recalled set from list j and its size
θ	Complete fitted parameter vector for one model

Table 7

Key for the six primary model columns in Table 8.

Code	Emotional-learning		Manuscript label
	multiplier	Early-LPP mapping	
A	Fixed at 1	None	Source-context baseline
B	Fixed at 1	General	General-LPP model
C	Fixed at 1	Emotion-dependent	Emotion-dependent- LPP model
D	Estimated	None	Preferential-learning model
E	Estimated	General	Preferential learning + General-LPP model
F	Estimated	Emotion-dependent	Preferential learning + Emotion- dependent-LPP model

Table 8

Best-found pooled parameter estimates for the six primary models. Values are rounded for presentation; square brackets identify fixed values. Parameter roles and fitted ranges are reported in Table 2.

Parameter	A	B	C	D	E	F
β_{enc}^T	0.86396	0.73472	0.75960	0.83243	0.00084	0.80907
β_{enc}^S	0.18153	0.02849	0.11016	0.04012	0.72584	0.08675
β_{start}	0.97321	0.98771	0.99999	0.99999	0.03122	0.99999
β_{rec}^T	0.70721	0.67079	0.14344	0.18358	0.05141	0.15337
β_{rec}^S	[1]	[1]	[1]	[1]	[1]	[1]
α	0.24130	3.53733	1.69400	3.81835	74.64477	1.85902
δ	68.959	28.732	39.126	18.541	48.979	27.334
γ	0.79003	0.46162	0.13427	0.25922	0.28439	0.21391
ϕ_s	28.721	0.48668	0.19390	0.02896	2.35576	0.22806
ϕ_d	2.20620	1.38064	4.43143	1.40662	2.11887	57.41072
τ_c	0.19046	2.17085	1.75392	3.32237	1.19079	1.57723
ω	0.22586	9.04693	1.38513	1.45162	1.03720	0.92996
ϕ_{emot}	[1]	[1]	[1]	1.50441	7.52498	1.94591
κ	[0]	0.05973	[0]	[0]	0.20799	[0]
κ_{emot}	[0]	[0]	0.18410	[0]	[0]	0.12805

Table 9

Exploratory Full-LPP evaluations. The original Full solutions are the best outputs from three differential-evolution starts. The embedded-parent candidates set $\kappa = 0$ and use the parameter vector from the corresponding Emotion-dependent model.

Preferential	NLL above			Best-		
emotional	Full-model	Original	Original	nested	Embedded-	evaluated
learning	p	Full NLL	Full AIC	model	parent NLL	Full AIC
Absent	13	8283.958	16593.916	3.265	8280.693	16587.387
Present	14	8241.604	16511.207	1.294	8240.310	16508.619

Table 10*Original parameter estimates from the two optimization-incomplete exploratory Full-LPP fits.*

Parameter	Emotional-learning multiplier	Emotional-learning multiplier
	fixed at 1	estimated
β_{enc}^T	0.63919	0.75269
β_{enc}^S	0.048073	0.090636
β_{start}	0.99928	0.99992
β_{rec}^T	0.19682	0.14939
β_{rec}^S	1 (fixed)	1 (fixed)
α	6.4214	7.5665
δ	29.362	61.357
γ	0.48335	0.26261
ϕ_s	0.29664	0.73550
ϕ_d	1.0567	0.85239
τ_c	3.1391	1.7813
ω	6.9497	3.5453
ϕ_{emot}	1 (fixed)	1.7823
κ	0.001684	0.006867
κ_{emot}	0.10562	0.11559

Figure 1

Observed benchmarks and predictions from four of the six pooled source-context models. Left: overall recall rates by category. Right: mean within-list centered Early LPP by memory status and category. Observed error bars are 95% participant-cluster bootstrap intervals. Predicted error bars are central 95% intervals across 200 simulated datasets and show simulation variability, not parameter-estimation uncertainty. Simulations used each list's observed recall count. The Early-LPP-by-memory-status pattern is a derived diagnostic, not a separately fitted target or out-of-sample validation.

